



Motifs Discovery Methods



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Outline

A practical guide to discovering recurring patterns (motifs) and rare subsequences (discords) using the `harbinger` package in R. Examples cover Matrix Profile–based methods and symbolic representations (SAX/XSAX).

01

Matrix Profile

STAMP, STOMP, SCRIMP, PMP, VALMOD

02

Symbolic Methods

SAX and XSAX for discretized patterns

03

Discord Analysis

Identifying anomalous subsequences

Notebook Structure

Notebooks are grouped by conceptual family to separate Matrix Profile methods from symbolic ones before comparing their results.

Matrix Profile (01-05)

- STAMP – approximate
- STOMP – exact
- SCRIMP – incremental
- PMP – multi-window
- VALMOD – variable length

Symbolic (06-07)

- SAX – standard alphabet
- XSAX – extended alphabet

Discords (08-09)

- Matrix Profile (STOMP)
- Symbolic discretization (SAX)

STAMP: Scalable Time Series Anytime Matrix Profile

The **Matrix Profile** is a powerful data structure that records, for each subsequence of a time series, the distance to its nearest neighbor. This enables efficient discovery of recurring patterns (motifs) and anomalies (discords).

Anytime Algorithm

Introduced by Yeh et al. (2016), STAMP approximates the Matrix Profile using random ordering of subsequence pairs. It can be interrupted at any time and still provide a valid result.

Key Properties

- Approximate results for speed
- Scalable for very long time series
- Employs a random sampling strategy
- Computational complexity: $O(n^2 \log n)$

When to Use

Ideal for large datasets where an exact Matrix Profile is computationally prohibitive, and an acceptable approximation is sufficient for timely analysis.

📄 **Citation:** Yeh, C.-C. M., et al. (2016). Matrix Profile I: All Pairs Similarity Joins for Time Series. IEEE ICDM.

01 – Motif Discovery with STAMP

STAMP uses random sampling to approximate the Matrix Profile, scaling well to long series. Each subsequence's distance to its nearest neighbor is computed.

```
# Carregar pacotes
library(daltoolbox)
library(harbinger)

# Carregar dados
data(examples_motifs)
dataset <- examples_motifs$simple

# Visualizar série
har_plot(harbinger(), dataset$serie)
```

```
# Definir modelo STAMP
# arg2: tamanho da janela
# arg3: número de motivos
model <- hmo_mp("stamp", 4, 3)

# Ajustar e detectar
model <- fit(model, dataset$serie)
detection <- detect(model, dataset$serie)

# Avaliar
evaluation <- evaluate(model,
  detection$event, dataset$event)
print(evaluation$confMatrix)
```

Step by Step

- Load `daltoolbox` and `harbinger`
- Select `examples_motifs$simple`
- Visualize the series before fitting the model
- Define `hmo_mp("stamp", 4, 3)`: window=4, 3 motifs
- Run `fit()` → `detect()` → `evaluate()`
- Result: 2 TP, 8 FP, 1 FN, 90 TN

📌 **Key Concept:** STAMP approximates the Matrix Profile via random sampling — efficient for long series, but does not guarantee an exact result.

STOMP: Exact Matrix Profile Computation

The STOMP algorithm provides a method for computing the Matrix Profile with guaranteed exactness, crucial for applications demanding high precision in time series analysis.

Exactness Guaranteed

Introduced by Zhu et al. (2016), STOMP (Scalable Time series Ordered-search Matrix Profile) computes the Matrix Profile exactly, delivering precise motif and discord discovery without approximation.

Computational Innovation

STOMP's key innovation exploits the relationship between consecutive dot products, avoiding redundant calculations and making it significantly faster than traditional brute-force approaches.

Key Properties

- Produces exact results.
- Uses sliding window dot product optimization.
- Time complexity: $O(n^2)$.
- Space complexity: $O(n)$.

STOMP vs. STAMP

While STAMP offers an anytime approximation, STOMP guarantees correctness. Both have similar practical runtimes for many datasets, but STOMP is preferred when precision is paramount.

When to Use

Ideal for scenarios where an exact Matrix Profile is indispensable and the dataset size is manageable enough to fit within available memory resources.

📄 **Citation:** Zhu, Y., et al. (2016). Matrix Profile II: Exploiting a Novel Algorithm and GPUs to Break the One Hundred Million Barrier for Time Series Motifs and Joins. IEEE ICDM.

02 – Motif Discovery with STOMP

STOMP is an exact and efficient algorithm for long series. Unlike STAMP, it guarantees the complete Matrix Profile without approximations, finishing in 0.01 sec on this example.

```
# Carregar pacotes e dados
library(daltoolbox)
library(harbinger)
data(examples_motifs)
dataset <- examples_motifs$simple

# Visualizar série bruta
har_plot(harbinger(), dataset$serie)
```

```
# Definir modelo STOMP
# arg2: tamanho da janela
# arg3: número de motivos
model <- hmo_mp("stomp", 4, 3)

# Ajustar e detectar
model <- fit(model, dataset$serie)
detection <- detect(model, dataset$serie)
# Finished in 0.01 secs

# Avaliar contra rótulos
evaluation <- evaluate(model,
  detection$event, dataset$event)
print(evaluation$confMatrix)
## TRUE FALSE
## TRUE  3   8
## FALSE 0  90
```

Step by step

- Same data structure as STAMP
- Simply change the argument to "stomp"
- STOMP detected 11 events (vs. 10 with STAMP)
- 3 TP, 8 FP, 0 FN — perfect recall on this dataset
- Execution time: 0.01 sec

❏ **Key concept:** STOMP computes the Matrix Profile exactly using a randomized column order — more precise than STAMP with similar computational cost.

SCRIMP: Incremental & Anytime Matrix Profile

The SCRIMP algorithm introduces an innovative anytime approach to Matrix Profile computation, enabling progressive refinement and valid approximations at any stage.

Anytime & Incremental

Introduced by Zhu et al. (2018), SCRIMP computes the Matrix Profile incrementally using random diagonal ordering, providing valid approximations at any stage.

Key Innovation

SCRIMP processes diagonals of the distance matrix in random order. This ensures that any partial result obtained is a valid, though incomplete, approximation of the full Matrix Profile.

Core Properties

- Anytime: interruptible with valid approximation.
- Random diagonal traversal for cache-efficiency.
- Converges to the exact result over time.
- Worst-case time complexity: $O(n^2)$.

SCRIMP vs. STAMP

While both are anytime algorithms, SCRIMP utilizes diagonal-based computation (more cache-friendly) instead of STAMP's column-based random sampling.

When to Use

Ideal for time-constrained scenarios requiring a result within a fixed budget, or for applications needing progressive refinement of the Matrix Profile.

📄 **Citation:** Zhu, Y., et al. (2018). Matrix Profile XI: SCRIMP++: Time Series Motif Discovery at Interactive Speeds. IEEE ICDM.

03 – Motif Discovery with SCRIMP

SCRIMP is an incremental (anytime) algorithm: it progressively approximates the Matrix Profile and can be interrupted at any point, refining the result as time allows.

```
# Carregar pacotes e dados
library(daltoolbox)
library(harbinger)
data(examples_motifs)
dataset <- examples_motifs$simple

# Visualizar série bruta
har_plot(harbinger(), dataset$serie)
```

```
# Definir modelo SCRIMP
model <- hmo_mp("scrimp", 4, 3)

# Ajustar e detectar
model <- fit(model, dataset$serie)
detection <- detect(model, dataset$serie)
# Finished in 0.03 secs

# Avaliar
evaluation <- evaluate(model,
  detection$event, dataset$event)
print(evaluation$confMatrix)
## TRUE FALSE
## TRUE  2   8
## FALSE 1  90
```

Step by step

- Replace the algorithm with "scrimp"
- Window=4, 3 motifs — same parameters
- Result identical to STAMP: 2 TP, 8 FP, 1 FN
- Time: 0.03 sec (slightly longer)
- Useful when execution time is constrained

❏ **Key concept:** SCRIMP is an *anytime* algorithm — it can be stopped early and still returns a valid approximation of the Matrix Profile.

PMP: The Pan Matrix Profile

The Pan Matrix Profile (PMP) algorithm extends the Matrix Profile to seamlessly analyze time series across multiple window sizes, providing a holistic view of patterns at varying scales.

01

Extended Concept

Introduced by Linardi et al. (2020), PMP simultaneously computes and consolidates Matrix Profiles for a range of window sizes, rather than just a single fixed one.

03

Hyperparameter Elimination

A major practical advantage of PMP is that it eliminates the need to manually tune the window size, often the most challenging hyperparameter in time series analysis.

02

Window-Size Agnostic

PMP reveals motifs that are inherently robust across different scales, offering higher precision and fewer false positives compared to single-window approaches.

04

Ideal Use Cases

Best suited for exploratory analysis where the optimal pattern scale is unknown, and when robust pattern discovery across diverse temporal resolutions is critical.

📄 **Citation:** Linardi, M., et al. (2020). VALMOD: A Suite for Easy and Exact Detection of Variable Length Motifs in Data Series. ACM SIGMOD.

04 – Motif Discovery with Pan Matrix Profile

PMP (Pan Matrix Profile) unifies matrix profiles across multiple window sizes, exposing motifs that persist regardless of the chosen window parameter.

```
# Carregar pacotes e dados
library(daltoolbox)
library(harbinger)
data(examples_motifs)
dataset <- examples_motifs$simple

# Visualizar série bruta
har_plot(harbinger(), dataset$serie)
```

```
# Definir modelo PMP
model <- hmo_mp("pmp", 4, 3)

# Ajustar e detectar
model <- fit(model, dataset$serie)
detection <- detect(model, dataset$serie)
# step: 1/1 binary idx: 1 window: 4

# Avaliar
evaluation <- evaluate(model,
  detection$event, dataset$event)
print(evaluation$confMatrix)
## TRUE FALSE
## TRUE  3   4
## FALSE 0  94
```

Step by step

- Use "pmp" as the first argument
- PMP detected only 7 events (more conservative)
- 3 TP, 4 FP, 0 FN — best precision among MP methods
- 94 TN — fewest false negatives
- Ideal when the optimal window size is unknown

📌 **Key concept:** PMP consolidates Matrix Profiles from multiple window sizes, revealing robust motifs that persist regardless of the chosen parameter.

VALMOD: Variable-Length Motif Discovery

The VALMOD algorithm extends Matrix Profile capabilities to uncover patterns of varying lengths, adapting to the inherent scales of motifs within time series data.

Core Concept

Introduced by Linardi et al. (2018), VALMOD directly addresses the challenge of discovering motifs across a range of subsequence lengths $[l_{min}, l_{max}]$, without requiring a fixed window size.

Efficient Search

It systematically searches for the optimal motif pair across all specified lengths, employing a lower-bounding technique to efficiently prune the search space and guarantee exact results.

Key Properties

- Supports variable-length subsequences directly.
- Enables incremental and streaming updates as new data arrives.
- Guarantees exact motif discovery within the specified length range.
- Utilizes length-normalized distance for robust comparisons.

VALMOD vs. PMP

Unlike PMP, which consolidates profiles for all window sizes, VALMOD focuses on finding the **single best** motif across a spectrum of lengths, emphasizing specific, high-quality pattern detection.

When to Use

- Streaming or online environments where data continuously flows.
- Scenarios where the optimal motif length is inherently unknown and requires automatic discovery.

 **Citation:** Linardi, M., et al. (2018). Matrix Profile X: VALMOD — Scalable Discovery of Variable-Length Motifs in Data Series. ACM SIGMOD.

05 – Motif Discovery with VALMOD

VALMOD maintains the Matrix Profile under streaming updates, enabling motif discovery in evolving series. It searches for recurring patterns across a range of subsequence lengths.

```
# Carregar pacotes e dados
library(daltoolbox)
library(harbinger)
data(examples_motifs)
dataset <- examples_motifs$simple

# Visualizar série bruta
har_plot(harbinger(), dataset$serie)
```

```
# Definir modelo VALMOD
# arg2: comprimento mínimo da janela
# arg3: número de motivos
model <- hmo_mp("valmod", 4, 3)

# Ajustar e detectar
model <- fit(model, dataset$serie)
detection <- detect(model, dataset$serie)
# max_number_motifs_found -1
# Finished in 0.02 secs

# Avaliar
evaluation <- evaluate(model,
  detection$event, dataset$event)
print(evaluation$confMatrix)
## TRUE FALSE
## TRUE  3  8
## FALSE 0 90
```

Step-by-step

- Use "valmod" as the algorithm
- The second argument defines the *minimum* window length
- Detected 11 events: 3 TP, 8 FP, 0 FN
- Same recall as STOMP, with streaming support
- The `max_number_motifs_found -1` message is expected

📌 **Key concept:** VALMOD supports variable-length subsequences and incremental updates — well-suited for real-time time series.

Summary: All Methods Covered

All methods follow the same pipeline in harbinger: `fit()` → `detect()` → `evaluate()`. The table below compares all nine notebooks by method family, strategy, and evaluation results on the `examples_motifs$simple` dataset (motifs) or `mitdb102` (discords).

Algorithm	Family	Strategy	Result (TP/FP/FN)	Time
STAMP	Matrix Profile	Random sampling (approx.)	2 / 8 / 1	0.02 sec
STOMP	Matrix Profile	Exact, randomized order	3 / 8 / 0	0.01 sec
SCRIMP	Matrix Profile	Incremental (anytime)	2 / 8 / 1	0.03 sec
PMP	Matrix Profile	Unified multi-window	3 / 4 / 0	—
VALMOD	Matrix Profile	Streaming, variable length	3 / 8 / 0	0.02 sec
SAX	Symbolic	Frequency of symbolic words	3 / 0 / 0	—
XSAX	Symbolic	Extended alphanumeric alphabet	3 / 0 / 0	—
STOMP (discord)	Discord · MP	Max nearest-neighbor distance	0 / 4 / 3	—
SAX (discord)	Discord · Symbolic	Rare symbolic words	0 / 121 / 3	—

Additional Sources

Explore the resources below to dive deeper into Harbinger and time series event detection.



CRAN Package

cran.r-project.org/web/packages/harbinger



Event Detection Resources

eic.cefet-rj.br/~eogasawara/en/event-detection-in-time-series



DAL Toolbox

cran.r-project.org/web/packages/daltoolbox



Issues & Discussions

github.com/cefet-rj-dal/harbinger/issues

Open questions, report bugs, or suggest improvements directly in the repository.

References

1. Yeh, C.-C. M., et al. (2016). *Matrix Profile I/II: All-pairs similarity joins and scalable time series motif/discord discovery*. IEEE ICDM.
2. Zhu, Y., et al. (2016). *Matrix Profile II: Exploiting a Novel Algorithm and GPUs to Break the One Hundred Million Barrier for Time Series Motifs and Joins*. IEEE ICDM.
3. Zhu, Y., et al. (2018). *Matrix Profile XI: SCRIMP++: Time Series Motif Discovery at Interactive Speeds*. IEEE ICDM.
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5. Tavenard, R., et al. (2020). *tsmp: The Matrix Profile in R*. The R Journal. doi:10.32614/RJ-2020-021
6. Lin, J., Keogh, E., Lonardi, S., Chiu, B. (2007). *A symbolic representation of time series, with implications for streaming algorithms*. Data Mining and Knowledge Discovery, 15, 107–144.
7. Keogh, E., Lin, J., Fu, A. (2005). *HOT SAX: Efficiently Finding the Most Unusual Time Series Subsequence*. IEEE ICDM.
8. Ogasawara, E., Salles, R., Porto, F., Pacitti, E. *Event Detection in Time Series*. Springer, 2025. doi:10.1007/978-3-031-75941-3